# Chapter 11 Notes

## Section 0

### Notes

Note that elementarily equivalence only requires pure formulas to be equivalent for the following reason: $A_1 \neq A_2$ may be true, and if it is, then if $a \in A_1$ and $a \notin A_2$, then there are formulas either true or false in $(A_1, R_1)$ (formulas involving $a$ as a constant) where truth in $(A_2, R_2)$ isn't even defined.

### Exercise 0.1

Suppose $\pi$ is an isomoprhism from $(A_1, R_1)$ to $(A_2, R_2)$. We show by complete induction on the degrees of formulas that for any pure formula $\phi(x_1, ..., x_n)$ and any elements $a_1, ..., a_n$ of $A_1$, the sentence $\phi(a_1, ..., a_n)$ is true in $(A_1, R_1)$ iff $\phi(\pi(a_1), ..., \pi(a_n))$ is true in $(A_2, R_2)$, and, for any pure sentence $\phi$, $\phi$ is true in $(A_1, R_1)$ iff $\phi$ is true in $(A_2, R_2)$.

In the base case, $Pab$ (for elements $a, b$ in $A_1$) is true in $(A_1, R_1)$ iff $aR_1b$ iff $\pi(a)R_2\pi(b)$ (because $\pi$ is an isomorphism) iff $P\pi(a)\pi(b)$ is true in $(A_2, R_2)$. Thus the base case holds.

Now suppose that the result holds for all formulas of degree less than $n$. That is, for any pure formula $\phi(x_1, ..., x_m)$ (of a degree less than $n$) and any elements $a_1, ..., a_m$ of $A_1$, $\phi(a_1, ..., a_n)$ is true in $(A_1, R_1)$ iff $\phi(\pi(a_1), ..., \pi(a_m))$ is true in $(A_2, R_2)$, and for any pure sentence $\phi$, $\phi$ is true in $(A_1, R_1)$ iff $\phi$ is true in $(A_2, R_2)$. On the one hand, we consider a pure formula with free variables of degree $n$. That is, we consider some pure formula $\phi(x_1, ..., x_m)$ and elements $a_1, ..., a_m$ of $A_1$. There are then multiple cases.

In the first case, $\phi(x_1, ..., x_m) = (\psi \wedge \gamma)$ for some pure formulas $\psi$ and $\gamma$ of degrees less than $n$. Either (1) neither $\psi$ nor $\gamma$ are sentences; (2) $\psi$ is a sentence and $\gamma$ is not; (3) $\psi$ is not a sentence and $\gamma$ is a sentence. We consider the first case - the other two are nearly identical. In that case, because neither $\psi$ nor $\gamma$ are sentences, $\psi = \psi(y_1, ..., y_k)$ for some $k < n$. Similarly $\gamma = \gamma(z_1, ..., z_l)$ for some $l < n$. Then $\phi(x_1, ..., x_m) = (\psi(y_1, ..., y_k) \wedge \gamma(z_1, ..., z_l))$. Then $\phi(a_1, ..., a_m) = (\psi(b_1, ..., b_k) \wedge \gamma(c_1, ..., c_l))$ where each $b_i = a_j$ for some $j$ (and each $c_i = a_j$ for some $j$). Then $\phi(a_1, ..., a_m)$ is true in $(A_1, R_1)$ iff $(\psi(b_1, ..., b_k) \wedge \gamma(c_1, ..., c_l))$ is true in $(A_1, R_1)$ iff $(\psi(\pi(b_1, ..., \pi(b_k))) \wedge \gamma(\pi(c_1), ..., \pi(c_l)))$ is true in $(A_2, R_2)$ (by the induction assumption) iff $\phi(\pi(a_1), ..., \pi(a_m))$ is true in $(A_2, R_2)$.

The second case, where $\phi(x_1, ..., x_m) = \sim\psi(x_1, ..., x_m)$ (for some formula $\psi(x_1, ..., x_m)$, is similar to the first case.

The third case is when $\phi(x_1, ..., x_m) = (\exists x)\phi(x, x_1, ..., x_m)$. Then $(\exists x)\phi(x, x_1, ..., x_m)$ is true iff $\phi(a, a_1, ..., a_m)$ is true for some $a \in A_1$, iff $\phi(\pi(a), \pi(a_1), ..., \pi(a_m))$ is true in $(A_2, R_2)$ iff $(\exists x)\phi(x, \pi(a_1), ..., \pi(a_m))$ is true in $(A_2, R_2)$.

On the other hand, we consider a pure sentence $\phi$ of degree $n$. The cases here are nearly identical to the above three cases - we give the case for existential quantification.

In that case, $\phi = (\exists x)\psi(x)$ for some $\psi$. Then $\phi$ is true in $(A_1, R_1)$ iff $\psi(a)$ is true in $(A_1, R_1)$ for some $a \in A_1$ iff $\psi(\pi(a))$ is true in $(A_2, R_2)$ (by the induction assumption) iff $(\exists x)\psi(x)$ is true in $(A_2, R_2)$.